WEEK 7: SQL Query Implementation for E-Commerce Database

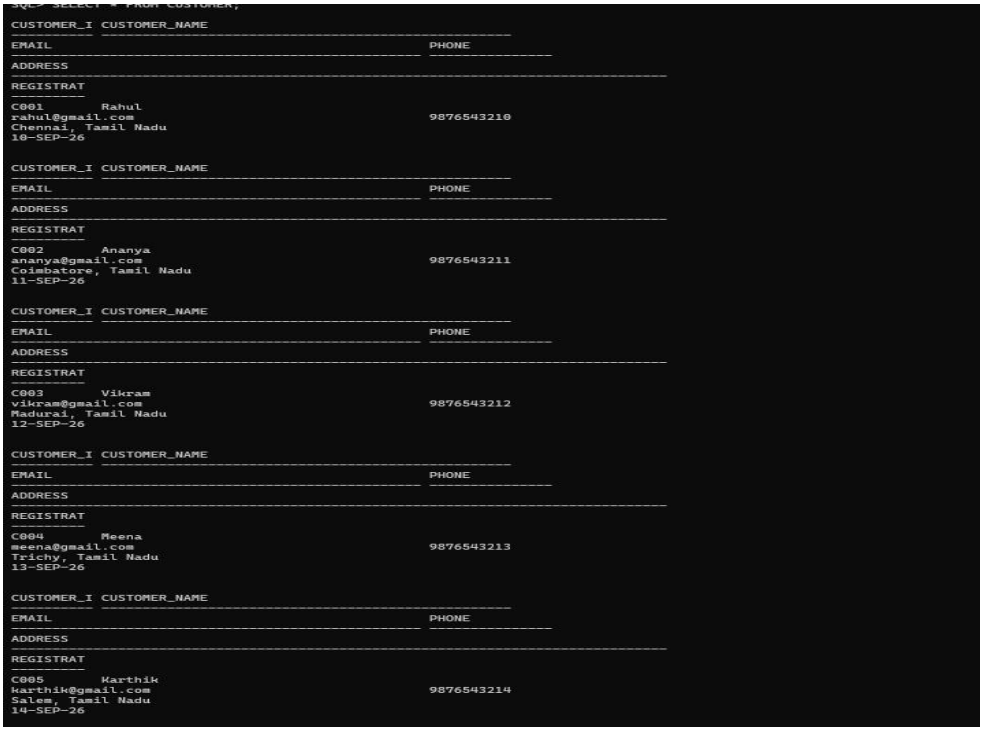
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Scenario:

The e-commerce company has collected customer, product, order, and inventory data. In this week, SQL queries are implemented to search, filter, sort, and generate reports from the existing Phone Case E-Commerce Database. The queries help retrieve customer and product information, identify product availability, analyze orders and payments, and generate basic business reports.

1. Perform basic SQL queries

1. Display all customer details.



```
SQL> SELECT * FROM CUSTOMER;

CUSTOMER_I CUSTOMER_NAME
---------- --------------------------------------------------
EMAIL                                              PHONE
-------------------------------------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
C001       Rahul
rahul@gmail.com                                    9876543210
Chennai, Tamil Nadu
10-SEP-26

CUSTOMER_I CUSTOMER_NAME
---------- --------------------------------------------------
EMAIL                                              PHONE
-------------------------------------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
C002       Ananya
ananya@gmail.com                                   9876543211
Coimbatore, Tamil Nadu
11-SEP-26

CUSTOMER_I CUSTOMER_NAME
---------- --------------------------------------------------
EMAIL                                              PHONE
-------------------------------------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
C003       Vikram
vikram@gmail.com                                   9876543212
Madurai, Tamil Nadu
12-SEP-26

CUSTOMER_I CUSTOMER_NAME
---------- --------------------------------------------------
EMAIL                                              PHONE
-------------------------------------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
C004       Meena
meena@gmail.com                                    9876543213
Trichy, Tamil Nadu
13-SEP-26

CUSTOMER_I CUSTOMER_NAME
---------- --------------------------------------------------
EMAIL                                              PHONE
-------------------------------------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
C005       Karthik
karthik@gmail.com                                  9876543214
Salem, Tamil Nadu
14-SEP-26
```

2. Display all available products.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE STOCK_QUANTITY > 0;

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
P001
iPhone Case
CAT001        499             50

P002
Samsung Case
CAT002        699             40

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------

P003
Redmi Case
CAT003        299             60

P004
OnePlus Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
CAT004        599             35

P005
Vivo Case
CAT005        799             25

P006

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
Realme Case
CAT001        399             45

P007
Oppo Case
CAT002        549             30


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
P008
Nothing Phone Case
CAT003        649             25

P009
Motorola Case
CAT004        449             35
```

```
P020

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
Vivo Shockproof Case
CAT005         499             35

P021
iPhone Printed Case
CAT001         699             25


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
P022
Samsung Printed Case
CAT002         649             27

P023
Redmi Printed Case
CAT003         399             48

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------

P024
OnePlus Printed Case
CAT004         599             33

P025
Vivo Printed Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
CAT005         449             40

P026
iPhone Matte Case
CAT001         749             24

P027

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
Samsung Matte Case
CAT002         699             29

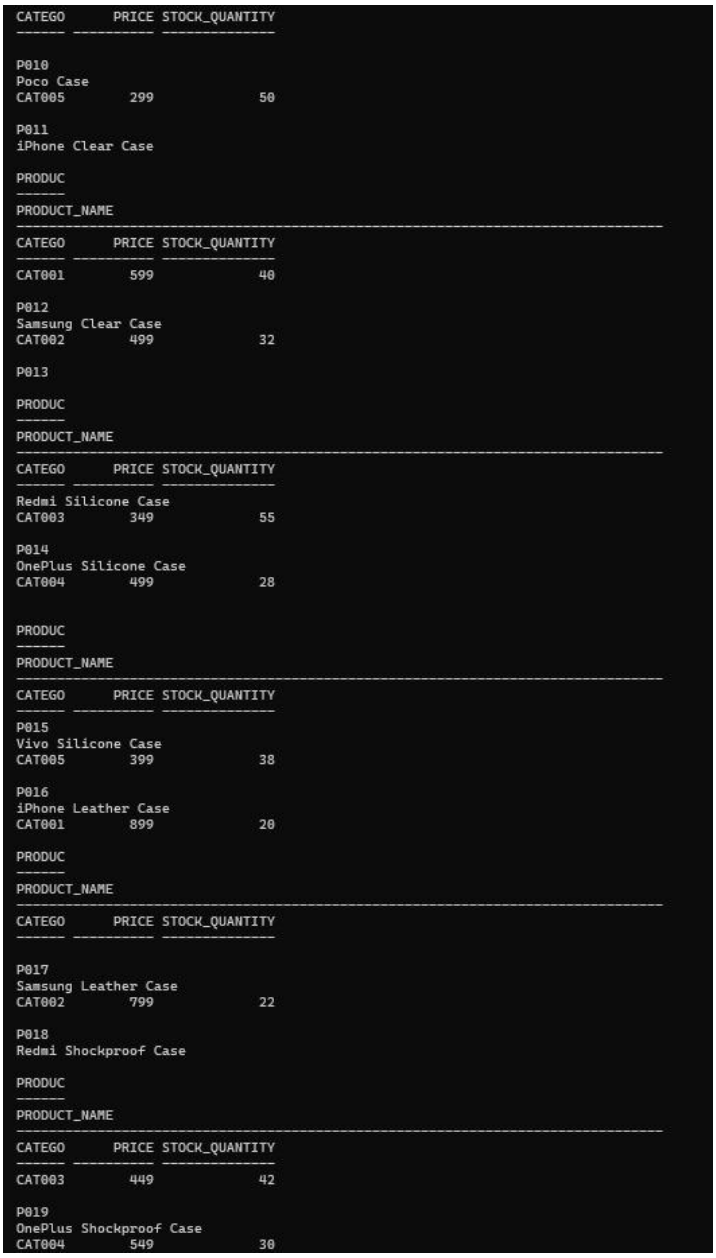
P028
Redmi Matte Case
CAT003         449             46


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
P029
OnePlus Matte Case
CAT004         649             31

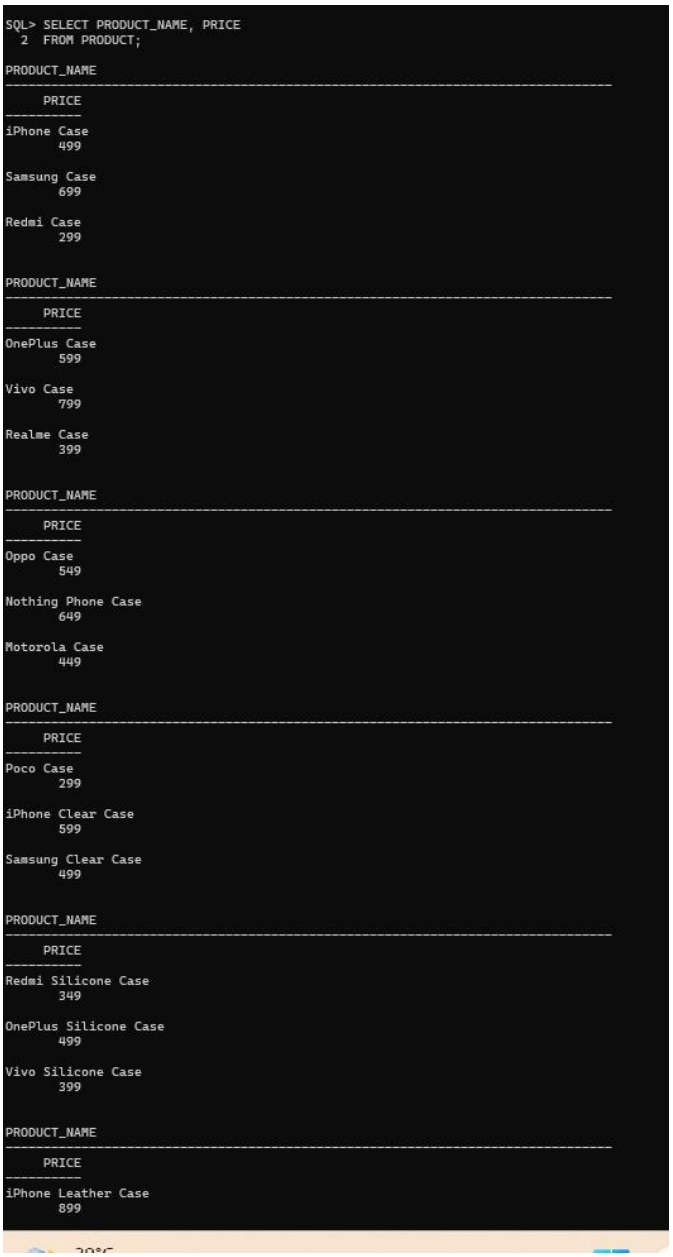
P030
Vivo Matte Case
CAT005         499             36

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------


30 rows selected.
```



3. Retrieve product names and prices only.



```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM PRODUCT;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
iPhone Case
       499

Samsung Case
       699

Redmi Case
       299


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
OnePlus Case
       599

Vivo Case
       799

Realme Case
       399


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Oppo Case
       549

Nothing Phone Case
       649

Motorola Case
       449


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Poco Case
       299

iPhone Clear Case
       599

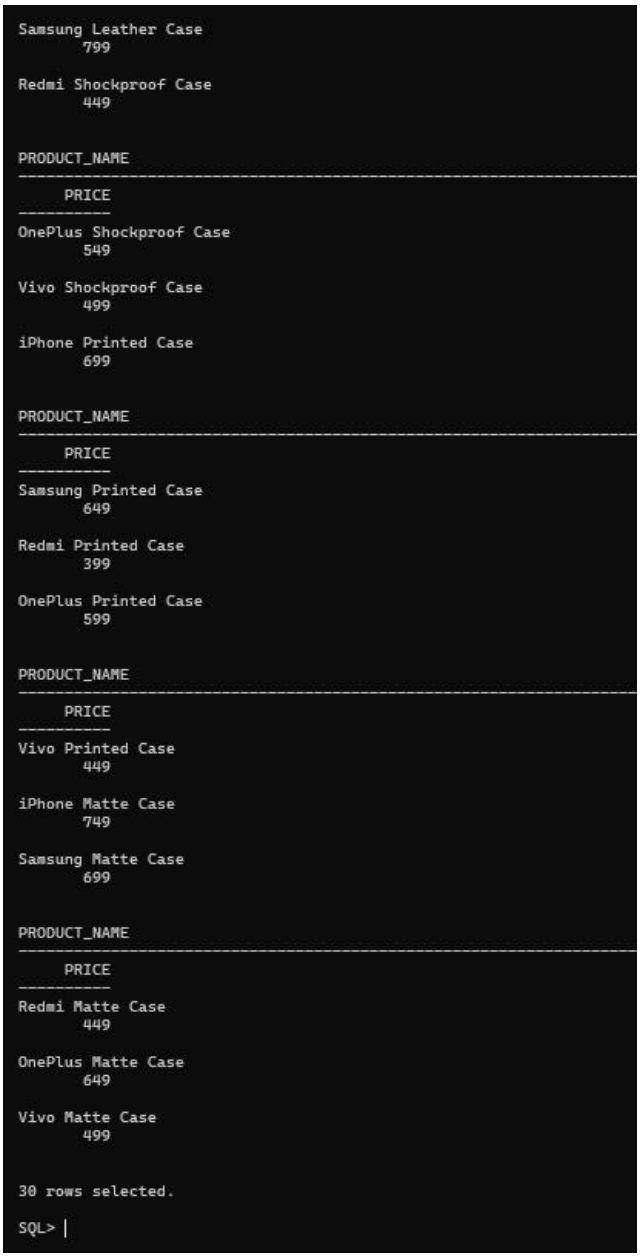
Samsung Clear Case
       499


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Redmi Silicone Case
       349

OnePlus Silicone Case
       499

Vivo Silicone Case
       399


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
iPhone Leather Case
       899
```



4. Display all orders placed by customers.

```
SQL> SELECT *
  2  FROM ORDERS;

  ORDER_ID CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ---------- --------- ------------ --------------------
       101 C001       15-SEP-26          499 Pending
       102 C002       16-SEP-26          699 Shipped
       103 C003       17-SEP-26          299 Delivered
       104 C004       18-SEP-26         1198 Pending
       105 C005       19-SEP-26          799 Shipped
       107 C002       21-SEP-26          599 Pending
       108 C003       22-SEP-26         1299 Shipped
       109 C004       23-SEP-26          749 Delivered
       110 C005       24-SEP-26          999 Pending
       111 C001       25-SEP-26          449 Shipped

10 rows selected.
```

5. Retrieve payment details

```
SQL> SELECT *
  2  FROM PAYMENT;

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
         1        101 15-SEP-26 UPI                  Success
               499

         2        102 16-SEP-26 Card                 Success
               699

         3        103 17-SEP-26 Cash                 Success
               299


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
         4        104 18-SEP-26 Net Banking          Pending
              1198

         5        105 19-SEP-26 UPI                  Success
               799

         6        107 21-SEP-26 Card                 Pending
               599


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
         7        108 22-SEP-26 UPI                  Success
              1299

         8        109 23-SEP-26 Cash                 Success
               749

         9        110 24-SEP-26 Net Banking          Pending
               999


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
        11        111 26-SEP-26 UPI                  Success
               599


10 rows selected.
```

2. Apply filtering conditions using WHERE

1. Find products with price greater than ₹5000.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE > 5000;

no rows selected

SQL>
```

2. Display products available in stock.

```
P020

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
Vivo Shockproof Case
CAT005         499             35

P021
iPhone Printed Case
CAT001         699             25


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
P022
Samsung Printed Case
CAT002         649             27

P023
Redmi Printed Case
CAT003         399             48

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------

P024
OnePlus Printed Case
CAT004         599             33

P025
Vivo Printed Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
CAT005         449             40

P026
iPhone Matte Case
CAT001         749             24

P027

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
Samsung Matte Case
CAT002         699             29

P028
Redmi Matte Case
CAT003         449             46


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------
P029
OnePlus Matte Case
CAT004         649             31

P030
Vivo Matte Case
CAT005         499             36

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE STOCK_QUANTITY
------ ---------- --------------


30 rows selected.
```

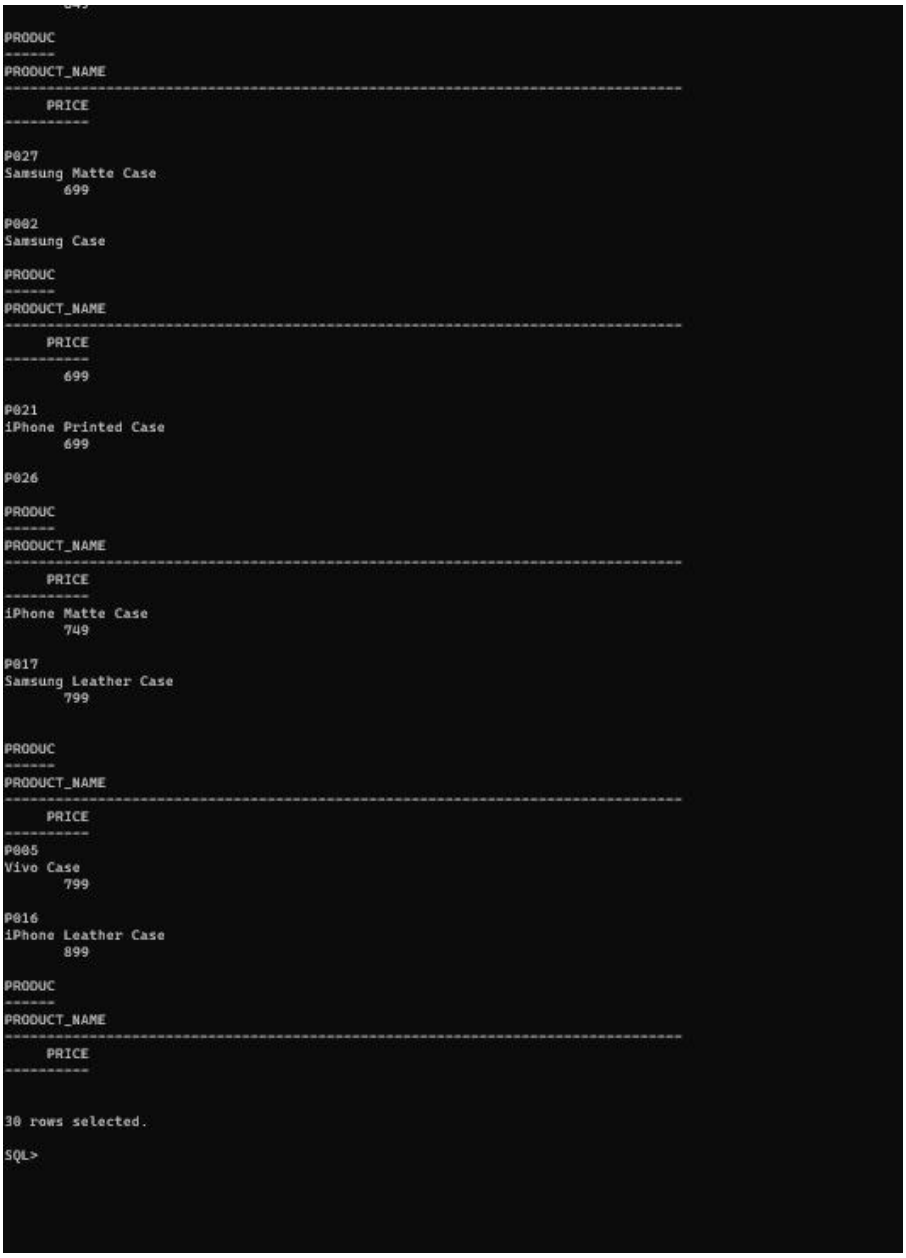
3. Display completed orders.

```
SQL> SELECT *
  2  FROM ORDERS
  3  WHERE UPPER(ORDER_STATUS) = 'DELIVERED';

  ORDER_ID CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ---------- --------- ------------ --------------------
       103 C003       17-SEP-26          299 Delivered
       109 C004       23-SEP-26          749 Delivered

SQL>
```

4. Find products with rating above 4.



3. Sort data using ORDER BY

1. Display products from lowest to highest price.

```
PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Vivo Shockproof Case
       499

P001
iPhone Case
       499


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P030
Vivo Matte Case
       499

P007
Oppo Case
       549

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

P019
OnePlus Shockproof Case
       549

P004
OnePlus Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       599

P024
OnePlus Printed Case
       599

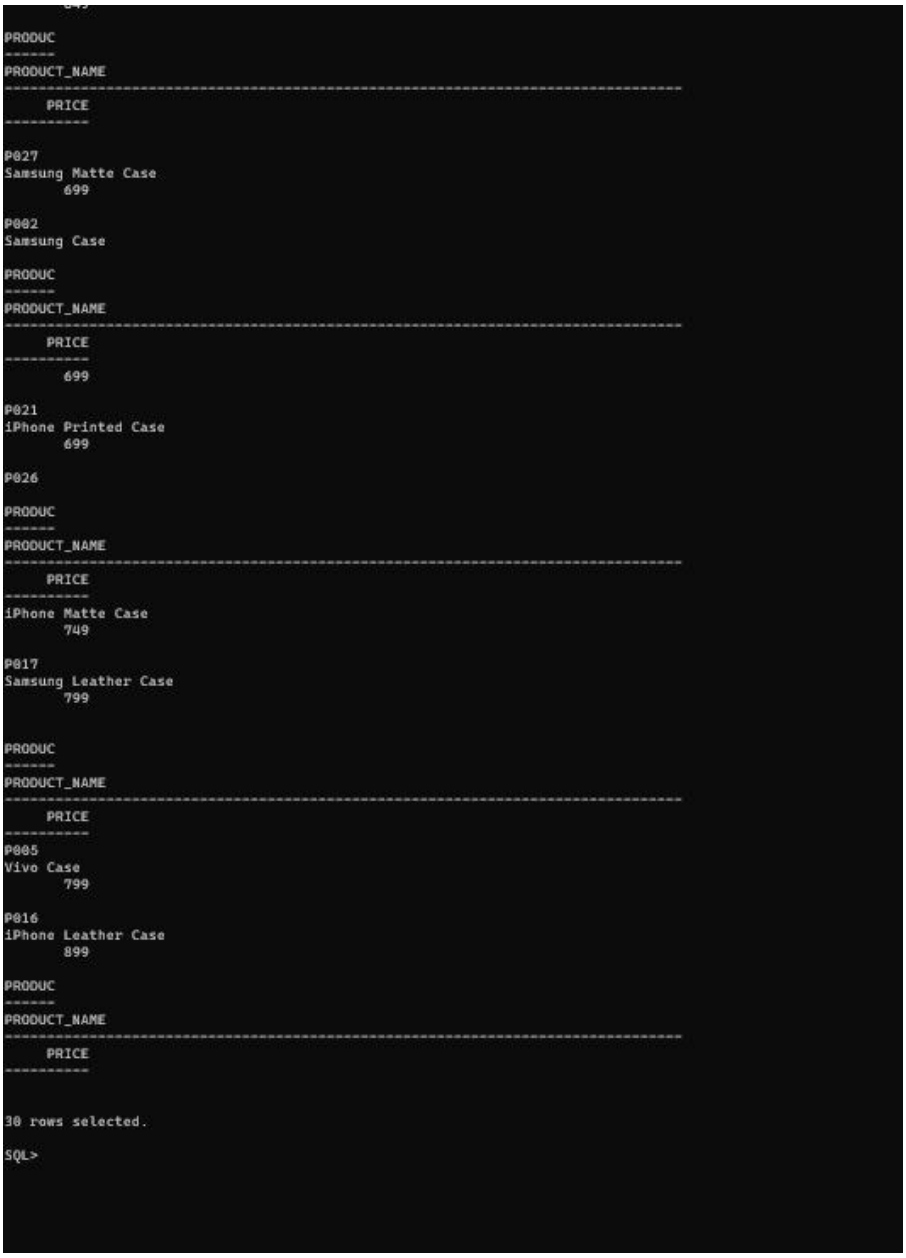
P011

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
iPhone Clear Case
       599

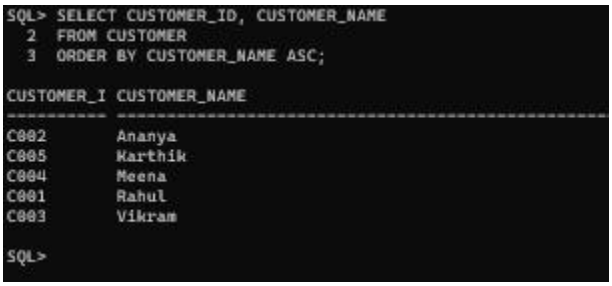
P022
Samsung Printed Case
       649


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P008
Nothing Phone Case
       649

P029
OnePlus Matte Case
       649
```



2. Display customers alphabetically.



3. Find the top 5 most expensive products.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM (
  3      SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  4      FROM PRODUCT
  5      ORDER BY PRICE DESC
  6  )
  7  WHERE ROWNUM <= 5;

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P016
iPhone Leather Case
       899

P005
Vivo Case
       799

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

P017
Samsung Leather Case
       799

P026
iPhone Matte Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       749

P002
Samsung Case
       699


SQL>
```

4. Display latest orders first.

```
SQL> SELECT *
  2  FROM ORDERS;

  ORDER_ID CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ---------- --------- ------------ --------------------
       101 C001       15-SEP-26          499 Pending
       102 C002       16-SEP-26          699 Shipped
       103 C003       17-SEP-26          299 Delivered
       104 C004       18-SEP-26         1198 Pending
       105 C005       19-SEP-26          799 Shipped
       107 C002       21-SEP-26          599 Pending
       108 C003       22-SEP-26         1299 Shipped
       109 C004       23-SEP-26          749 Delivered
       110 C005       24-SEP-26          999 Pending
       111 C001       25-SEP-26          449 Shipped

10 rows selected.
```

4. Retrieve unique values using DISTINCT

1. Display unique product categories.

```
SQL> SELECT DISTINCT CATEGORY_ID
  2  FROM PRODUCT;

CATEGO
------
CAT003
CAT002
CAT004
CAT001
CAT005

SQL>
```

2. Find different payment methods used by customers.

```
SQL> SELECT DISTINCT PAYMENT_MODE
  2  FROM PAYMENT;

PAYMENT_MODE
--------------------
Net Banking
Card
Cash
UPI

SQL>
```

5. Search products based on conditions

1. Find products within a specific price range (₹20,000 to ₹40,000).

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE BETWEEN 20000 AND 40000;

no rows selected

SQL>
```

Phone cases do not have price range of 20,000 to 40,000 So, 500 to 600

```
Connected.
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE BETWEEN 500 AND 600;

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P004
OnePlus Case
       599


SQL>
```

2. Display products belonging to a particular category.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, CATEGORY_ID, PRICE
  2  FROM PRODUCT
  3  WHERE CATEGORY_ID = 'CAT001';

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE
------ ----------
P001
iPhone Case
CAT001         499

P006
Realme Case
CAT001         399

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE
------ ----------

P011
iPhone Clear Case
CAT001         599

P016
iPhone Leather Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE
------ ----------
CAT001         899

P021
iPhone Printed Case
CAT001         699

P026

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGO      PRICE
------ ----------
iPhone Matte Case
CAT001         749


6 rows selected.
```

3. Find products currently available in inventory.

```
SQL> SELECT * FROM Inventory
  2  WHERE Stock_Status = 'Available';

INVENTO PRODUC SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------ ------- -------------- ----------- ---------
I000001 P001   S000002             50 Available   26-OCT-26
I000002 P002   S000002             40 Available   26-OCT-26
I000003 P003   S000002             35 Available   26-OCT-26
I000004 P004   S000002             25 Available   26-OCT-26
I000005 P005   S000002             30 Available   26-OCT-26
```

4. Search products using product name.

```
SQL> SELECT * FROM Product WHERE Product_Name = 'Luxury Gold Case';

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P000022
Luxury Gold Case
CAT0005        699             12
```

5. Find low-stock products

```
SQL> SELECT * FROM Inventory
  2  WHERE Stock_Quantity < 30;

INVENTO PRODUC SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------ ------- -------------- ----------- ---------
I000004 P004   S000002             25 Available   26-OCT-26

SQL>
```

6. Retrieve customer and product information 1. Display customer details with their orders.

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, O.ORDER_ID,
  2         O.ORDER_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS
  3  FROM Customer C
  4  INNER JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  ORDER BY O.ORDER_ID;

CUSTOMER_I CUSTOMER_NAME                                     ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
C001       Rahul                                                 101
15-SEP-26          499 Pending

C002       Ananya                                                102
16-SEP-26          699 Shipped

C003       Vikram                                                103
17-SEP-26          299 Delivered


CUSTOMER_I CUSTOMER_NAME                                     ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
C004       Meena                                                 104
18-SEP-26         1198 Pending

C005       Karthik                                               105
19-SEP-26          799 Shipped

C002       Ananya                                                107
21-SEP-26          599 Pending

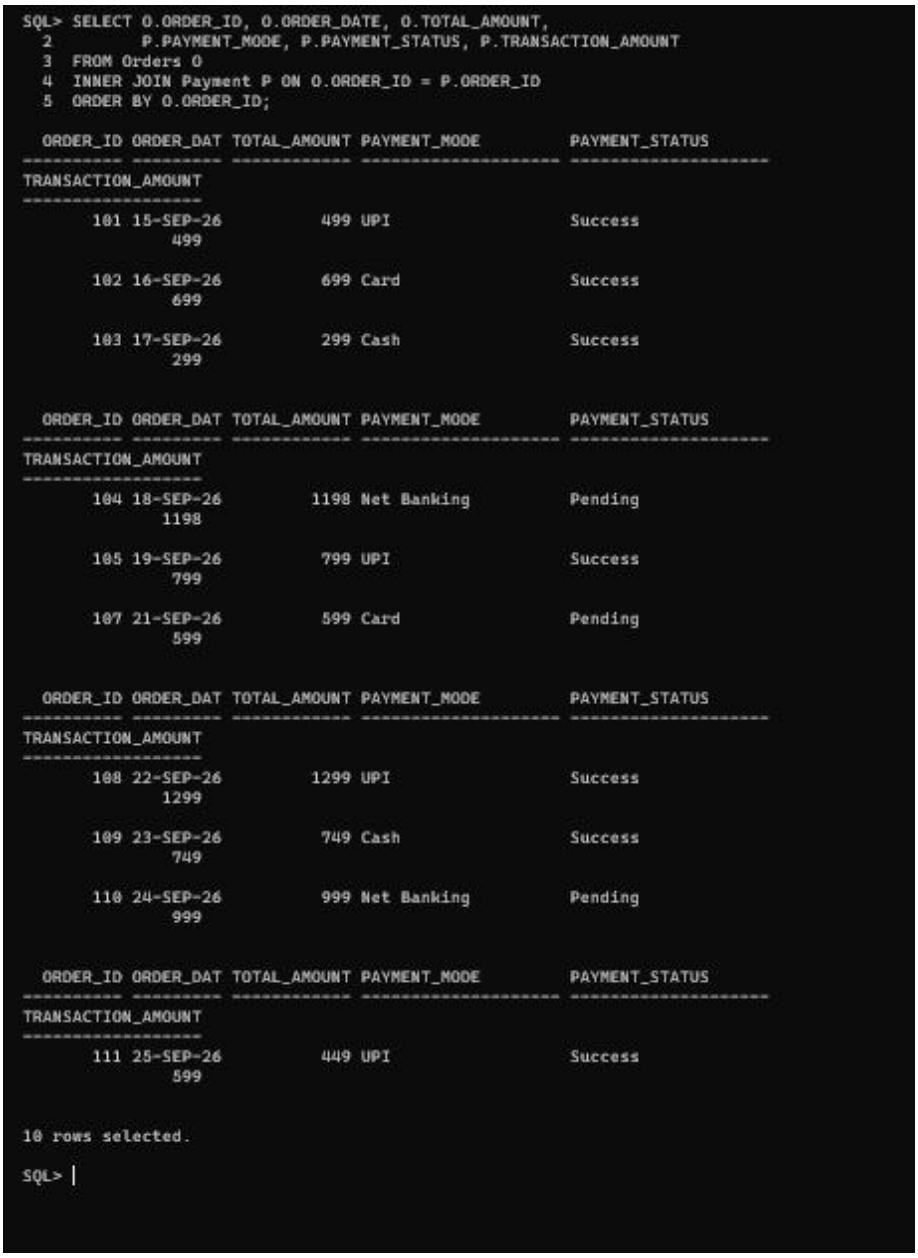

CUSTOMER_I CUSTOMER_NAME                                     ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
C003       Vikram                                                108
22-SEP-26         1299 Shipped

C004       Meena                                                 109
23-SEP-26          749 Delivered

C005       Karthik                                               110
24-SEP-26          999 Pending


CUSTOMER_I CUSTOMER_NAME                                     ORDER_ID
---------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
C001       Rahul                                                 111
25-SEP-26          449 Shipped


10 rows selected.
```



```
SQL> SELECT O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT,
  2         P.PAYMENT_MODE, P.PAYMENT_STATUS, P.TRANSACTION_AMOUNT
  3  FROM Orders O
  4  INNER JOIN Payment P ON O.ORDER_ID = P.ORDER_ID
  5  ORDER BY O.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ ------------------- --------------------
TRANSACTION_AMOUNT
------------------
       101 15-SEP-26          499 UPI                 Success
               499

       102 16-SEP-26          699 Card                Success
               699

       103 17-SEP-26          299 Cash                Success
               299


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ ------------------- --------------------
TRANSACTION_AMOUNT
------------------
       104 18-SEP-26         1198 Net Banking         Pending
              1198

       105 19-SEP-26          799 UPI                 Success
               799

       107 21-SEP-26          599 Card                Pending
               599


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ ------------------- --------------------
TRANSACTION_AMOUNT
------------------
       108 22-SEP-26         1299 UPI                 Success
              1299

       109 23-SEP-26          749 Cash                Success
               749

       110 24-SEP-26          999 Net Banking         Pending
               999


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ ------------------- --------------------
TRANSACTION_AMOUNT
------------------
       111 25-SEP-26          449 UPI                 Success
               599


10 rows selected.

SQL>
```

2. Display product details with category information.

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID, P.PRODUCT_NAME,
  2         OD.QUANTITY, OD.PRICE
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  JOIN Order_Details OD ON O.ORDER_ID = OD.ORDER_ID
  6  JOIN Product P ON OD.PRODUCT_ID = P.PRODUCT_ID
  7  ORDER BY C.CUSTOMER_NAME, O.ORDER_ID;

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Ananya                                                102
Samsung Case
         1        699

Ananya                                                107
Samsung Case
         3        599

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

Karthik                                               105
Vivo Case
         1        950

Karthik                                               110
Vivo Case

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         2        999

Meena                                                 104
OnePlus Case
         2        599

Meena                                                 109

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
OnePlus Case
         1        749

Rahul                                                 101
iPhone Case
         1        499


CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Rahul                                                 111
iPhone Case
         1        449

Vikram                                                103
Redmi Case
         1        299

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

Vikram                                                108
Redmi Case
         2        299

10 rows selected.
```

3. Find customers who purchased a specific product.

```
SQL> SELECT DISTINCT C.CUSTOMER_ID, C.CUSTOMER_NAME
  2  FROM CUSTOMER C
  3  JOIN ORDERS O
  4  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  JOIN ORDER_DETAILS OD
  6  ON O.ORDER_ID = OD.ORDER_ID
  7  WHERE OD.PRODUCT_ID = 'P001';

CUSTOMER_I CUSTOMER_NAME
---------- --------------------------------------------------
C001       Rahul
```

4. Display products purchased by each customer

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID,
  2         P.PRODUCT_NAME, OD.QUANTITY, OD.PRICE
  3  FROM CUSTOMER C
  4  JOIN ORDERS O
  5  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  6  JOIN ORDER_DETAILS OD
  7  ON O.ORDER_ID = OD.ORDER_ID
  8  JOIN PRODUCT P
  9  ON OD.PRODUCT_ID = P.PRODUCT_ID
 10  ORDER BY C.CUSTOMER_NAME, O.ORDER_ID;

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Ananya                                                102
Samsung Case
         1        699

Ananya                                                107
Samsung Case
         3        599

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

Karthik                                               105
Vivo Case
         1        950

Karthik                                               110
Vivo Case

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         2        999

Meena                                                 104
OnePlus Case
         2        599

Meena                                                 109

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
OnePlus Case
         1        749

Rahul                                                 101
iPhone Case
         1        499
```

```
CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Rahul                                                 111
iPhone Case
         1        449

Vikram                                                103
Redmi Case
         1        299

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

Vikram                                                108
Redmi Case
         2        299


10 rows selected.
```

7. Apply multiple filtering conditions 1. Find phone case products costing more than ₹600.

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, P.PRICE
  2  FROM PRODUCT P
  3  JOIN CATEGORY C
  4  ON P.CATEGORY_ID = C.CATEGORY_ID
  5  WHERE P.PRICE > 500;

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P002
Samsung Case
       699

P004
OnePlus Case
       599
```

2. Search products containing the word “CLEAR”.

```
SQL> SELECT PRODUCT_NAME, PRICE, STOCK_QUANTITY,
  2         CASE
  3             WHEN STOCK_QUANTITY > 0 THEN 'Available'
  4             ELSE 'Out of Stock'
  5         END AS AVAILABILITY_STATUS
  6  FROM PRODUCT
  7  ORDER BY PRODUCT_NAME;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Motorola Case
       449             35 Available

Nothing Phone Case
       649             25 Available

OnePlus Case
       599             35 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
OnePlus Matte Case
       649             31 Available

OnePlus Printed Case
       599             33 Available

OnePlus Shockproof Case
       549             30 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
OnePlus Silicone Case
       499             28 Available

Oppo Case
       549             30 Available

Poco Case
       299             50 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Realme Case
       399             45 Available

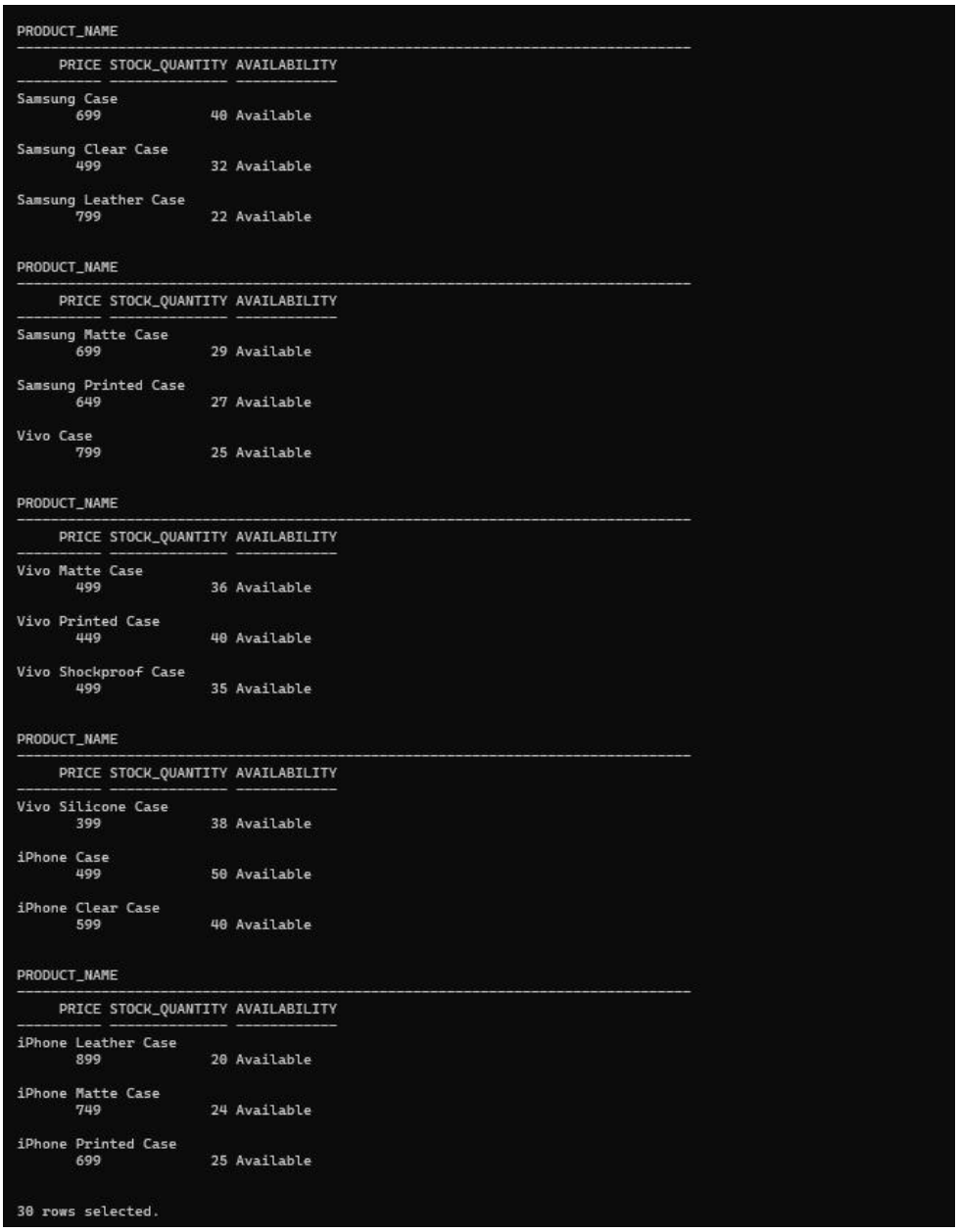
Redmi Case
       299             60 Available

Redmi Matte Case
       449             46 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Redmi Printed Case
       399             48 Available

Redmi Shockproof Case
       449             42 Available

Redmi Silicone Case
       349             55 Available
```



```
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Samsung Case
       699             40 Available

Samsung Clear Case
       499             32 Available

Samsung Leather Case
       799             22 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Samsung Matte Case
       699             29 Available

Samsung Printed Case
       649             27 Available

Vivo Case
       799             25 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Vivo Matte Case
       499             36 Available

Vivo Printed Case
       449             40 Available

Vivo Shockproof Case
       499             35 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Vivo Silicone Case
       399             38 Available

iPhone Case
       499             50 Available

iPhone Clear Case
       599             40 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
iPhone Leather Case
       899             20 Available

iPhone Matte Case
       749             24 Available

iPhone Printed Case
       699             25 Available


30 rows selected.
```

3. Find orders placed within a specific date range.

```
SQL> SELECT ORDER_ID, CUSTOMER_ID, ORDER_DATE,
  2         TOTAL_AMOUNT, ORDER_STATUS
  3  FROM ORDERS
  4  WHERE ORDER_DATE >= DATE '2026-09-20'
  5    AND ORDER_DATE < DATE '2026-10-01'
  6  ORDER BY ORDER_DATE;

  ORDER_ID CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ---------- --------- ------------ --------------------
       107 C002       21-SEP-26          599 Pending
       108 C003       22-SEP-26         1299 Shipped
       109 C004       23-SEP-26          749 Delivered
       110 C005       24-SEP-26          999 Pending
       111 C001       25-SEP-26          449 Shipped

SQL>
```

8. Generate basic business reports Report

1: Product Availability Report Display product name, price, stock quantity, and availability status.

```
SQL> SELECT PRODUCT_NAME, PRICE, STOCK_QUANTITY,
  2         CASE
  3             WHEN STOCK_QUANTITY > 0 THEN 'Available'
  4             ELSE 'Out of Stock'
  5         END AS AVAILABILITY_STATUS
  6  FROM PRODUCT
  7  ORDER BY PRODUCT_NAME;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Motorola Case
       449             35 Available

Nothing Phone Case
       649             25 Available

OnePlus Case
       599             35 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
OnePlus Matte Case
       649             31 Available

OnePlus Printed Case
       599             33 Available

OnePlus Shockproof Case
       549             30 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
OnePlus Silicone Case
       499             28 Available

Oppo Case
       549             30 Available

Poco Case
       299             50 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Realme Case
       399             45 Available

Redmi Case
       299             60 Available

Redmi Matte Case
       449             46 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Redmi Printed Case
       399             48 Available

Redmi Shockproof Case
       449             42 Available

Redmi Silicone Case
       349             55 Available
```

```
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Samsung Case
       699             40 Available

Samsung Clear Case
       499             32 Available

Samsung Leather Case
       799             22 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Samsung Matte Case
       699             29 Available

Samsung Printed Case
       649             27 Available

Vivo Case
       799             25 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Vivo Matte Case
       499             36 Available

Vivo Printed Case
       449             40 Available

Vivo Shockproof Case
       499             35 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Vivo Silicone Case
       399             38 Available

iPhone Case
       499             50 Available

iPhone Clear Case
       599             40 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
iPhone Leather Case
       899             20 Available

iPhone Matte Case
       749             24 Available

iPhone Printed Case
       699             25 Available


30 rows selected.
```

Report 2: Customer Report

a. Total number of customers

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS
  2  FROM CUSTOMER;

TOTAL_CUSTOMERS
---------------
              5
```

b. New customer registrations

```
SQL> SELECT CUSTOMER_ID, CUSTOMER_NAME, REGISTRATION_DATE
  2  FROM CUSTOMER
  3  ORDER BY REGISTRATION_DATE DESC;

CUSTOMER_I CUSTOMER_NAME                                          REGISTRAT
---------- ---------------------------------------------------- ---------
C005       Karthik                                              14-SEP-26
C004       Meena                                                13-SEP-26
C003       Vikram                                               12-SEP-26
C002       Ananya                                               11-SEP-26
C001       Rahul                                                10-SEP-26

SQL> |
```

Report 3: Order Report a. Total orders

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
          10
```

b. Completed orders

```
SQL> SELECT COUNT(*) AS COMPLETED_ORDERS
  2  FROM ORDERS
  3  WHERE UPPER(ORDER_STATUS) = 'COMPLETED';

COMPLETED_ORDERS
----------------
               0
```

C. Pending orders

```
SQL> SELECT COUNT(*) AS PENDING_ORDERS
  2  FROM ORDERS
  3  WHERE UPPER(ORDER_STATUS) = 'PENDING';

PENDING_ORDERS
--------------
             4
```

d. Cancelled orders

```
SQL> SELECT COUNT(*) AS CANCELLED_ORDERS
  2  FROM ORDERS
  3  WHERE UPPER(ORDER_STATUS) = 'CANCELLED';

CANCELLED_ORDERS
----------------
               0

SQL> |
```

Report 4:

Product Performance Report a. Highest-priced products

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE = (SELECT MAX(PRICE) FROM PRODUCT);

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P016
iPhone Leather Case
       899
```

b. Most-reviewed products

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME,
  2         COUNT(R.REVIEW_ID) AS TOTAL_REVIEWS
  3  FROM PRODUCT P
  4  JOIN REVIEW R
  5  ON P.PRODUCT_ID = R.PRODUCT_ID
  6  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  7  ORDER BY TOTAL_REVIEWS DESC;

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_REVIEWS
-------------
P001
iPhone Case
           8

P002
Samsung Case
           6

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_REVIEWS
-------------
```

c. Available products

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, STOCK_QUANTITY
  2  FROM PRODUCT
  3  WHERE STOCK_QUANTITY > 0
  4  ORDER BY PRODUCT_NAME;

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P009
Motorola Case
       449             35

P008
Nothing Phone Case
       649             25

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P004
OnePlus Case
       599             35

P029
OnePlus Matte Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       649             31

P024
OnePlus Printed Case
       599             33

P019

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
OnePlus Shockproof Case
       549             30

P014
OnePlus Silicone Case
       499             28


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P007
Oppo Case
       549             30

P010
Poco Case
       299             50
```

```
P030
Vivo Matte Case
       499             36

P025
Vivo Printed Case
       449             40

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P020
Vivo Shockproof Case
       499             35

P015
Vivo Silicone Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       399             38

P001
iPhone Case
       499             50

P011

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
iPhone Clear Case
       599             40

P016
iPhone Leather Case
       899             20


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P026
iPhone Matte Case
       749             24

P021
iPhone Printed Case
       699             25

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------


30 rows selected.
```

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, STOCK_QUANTITY
  2  FROM PRODUCT
  3  WHERE STOCK_QUANTITY > 0
  4  ORDER BY PRODUCT_NAME;

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P009
Motorola Case
       449             35

P008
Nothing Phone Case
       649             25

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P004
OnePlus Case
       599             35

P029
OnePlus Matte Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       649             31

P024
OnePlus Printed Case
       599             33

P019

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
OnePlus Shockproof Case
       549             30

P014
OnePlus Silicone Case
       499             28


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P007
Oppo Case
       549             30

P010
Poco Case
       299             50

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P006
Realme Case
       399             45

P003
Redmi Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
```

```
     PRICE STOCK_QUANTITY
---------- --------------
Samsung Printed Case
       649             27

P005
Vivo Case
       799             25


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P030
Vivo Matte Case
       499             36

P025
Vivo Printed Case
       449             40

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P020
Vivo Shockproof Case
       499             35

P015
Vivo Silicone Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       399             38

P001
iPhone Case
       499             50

P011

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
iPhone Clear Case
       599             40

P016
iPhone Leather Case
       899             20


PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P026
iPhone Matte Case
       749             24

P021
iPhone Printed Case
       699             25

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------


30 rows selected.

SQL>
```